

The Nearest Target Is the Wrong One: Target Separation in Arc2Face Identity Unlearning

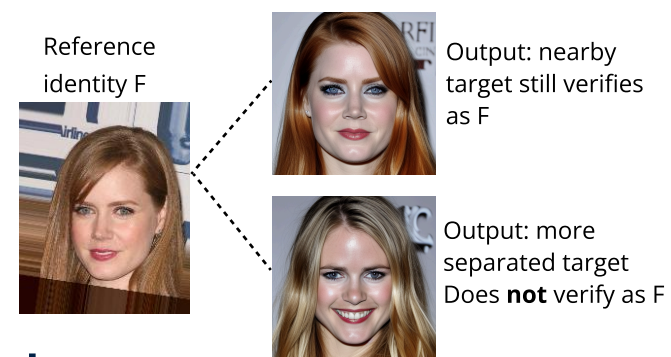
Zeynel Tok

University of Oxford · zeynel.tok@cs.ox.ac.uk

1. Motivation

A changed output can still verify as the original identity.

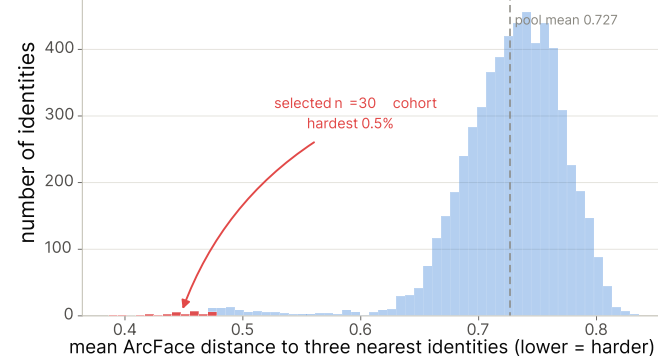
Identity conditioned generators can reproduce a person's likeness from a few references [1,2], making identity unlearning privacy-relevant. Visibly changed outputs may still retain residual information [3,4], so forgetting must be verified in recognition space not judged by appearance alone.



Does greater source-target separation improve forgetting without harming retained identities?

2. Experimental Setup

Testing the most locally entangled identities.



From 5,956 identities, we select the 30 whose three nearest ArcFace neighbours have the smallest mean distance: the most locally entangled 0.5%.

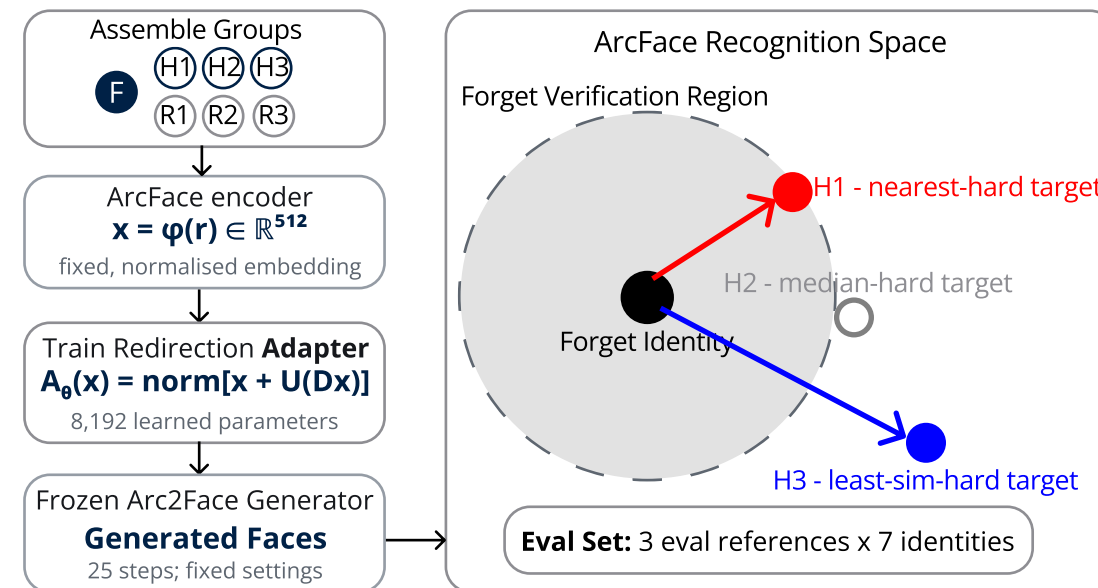
This is a stress test, not an estimate of average case performance.



The hard neighbours serve as both challenging retain identities and the shared target candidate set, the random identities test identity preservation.

3. Main Idea

Identity-pathway redirection: Target choice controls forgetting.



Across policies: the group, candidate set, adapter architecture, objective, training budget and frozen generator are fixed. Only the target selection rule changes.

Nearest-Hard Target

The generated image remains too close to the source region, so generated faces can still match the Forget Identity.

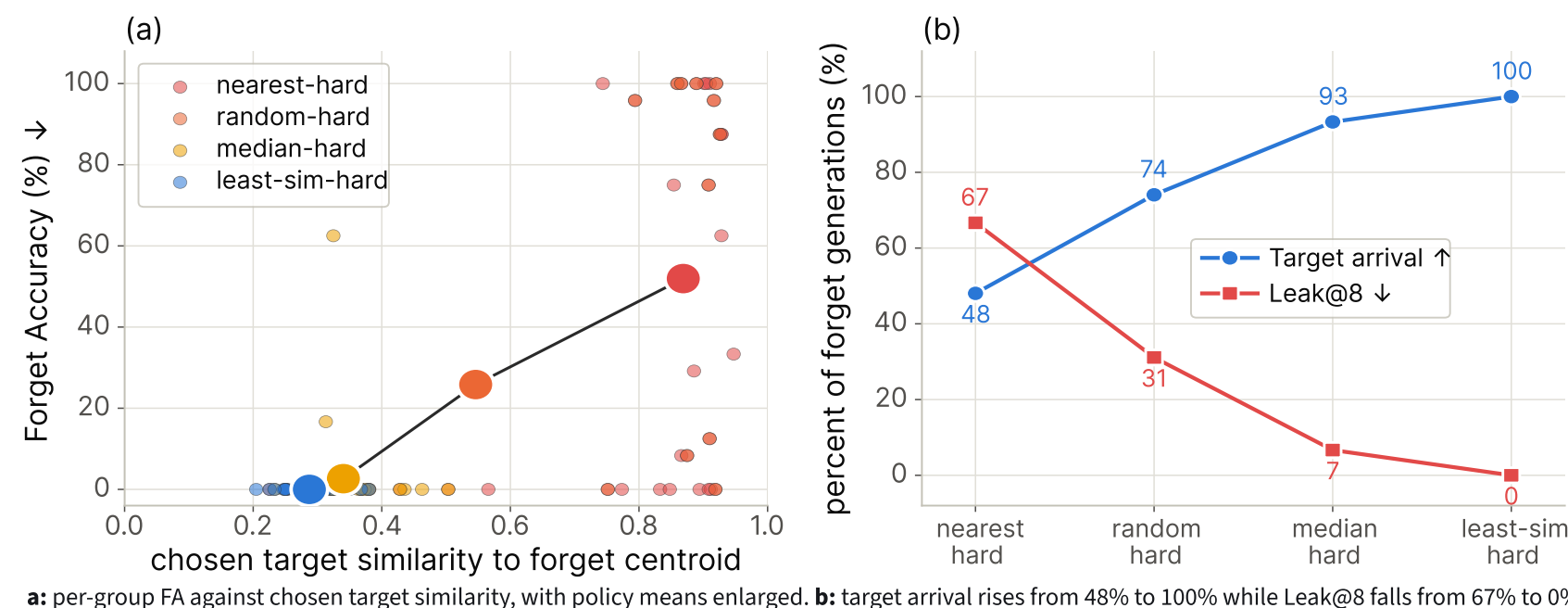
Least-Sim-Hard Target

With least-sim-hard, all detected forget-conditioned outputs verify as the selected target, and none verify as F.

Adapter Objective:

- Forget:** Redirect F toward the selected target H
 - Retain:** Keep the six retain embeddings near their originals
 - Preserve:** Keep 512 background embeddings stable at every step
- The same adapter is applied to forget and retain inputs, without identity-specific routing.

4. Primary Result



a: per-group FA against chosen target similarity, with policy means enlarged. **b:** target arrival rises from 48% to 100% while Leak@8 falls from 67% to 0%. Intermediate policies support this trend.

- Forget Accuracy (FA):** Output is still recognised as the identity that needs to be forgotten.
- Retain Accuracy (RA):** Retained output recognised as its intended identity.
- Leak@8:** For each fit reference, did any of its eight outputs still verify as F? Reported as the percentage of references that leak.
- Target Arrival:** Output image recognised as the selected target.

Policy	Target Similarity	FA ↓	RA ↑	Leak@8 ↓	# FA=0
nearest-hard	.8689	51.94	88.79	66.67	9/30
random-hard	.5460	25.83	88.86	31.11	20/30
median-hard	.3403	2.64	88.75	6.67	28/30
least-sim-hard	.2874	0.00	88.86	0.00	30/30

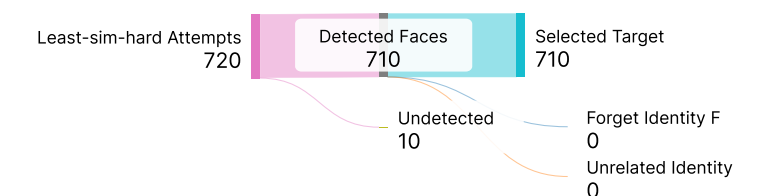
5. Findings & Conclusion

1. Retention is stable across policies

Retention Target	Retain Accuracy (RA) - Per Policy
All 6 Retains (mean)	88.75-88.86%
3 Random Retains	97.80-97.89%
3 Hard Neighbours	79.78-79.96%
Selected Target (1 of 4)	53.9 → 77.1 → 92.2 → 94.3%

Overall retention changes little across policies. Hard neighbours remain more difficult than random retains, while RA for the selected target is higher under policies choosing less similar targets.

2. Redirection not Rejection



All 710 detected outputs verify as the selected target. Low FA therefore reflects successful target arrival rather than outputs becoming undetectable or drifting to unrelated identities.

3. The Trend holds under a separate image recogniser

Policy	ArcFace FA	AdaFace FA
nearest-hard	51.94	49.52
random-hard	25.83	22.86
median-hard	2.64	1.94
least-sim-hard	0.00	0.00

A separately trained AdaFace recogniser reproduces the same ordering and reaches zero forget accuracy under least-sim-hard.

Scope:

The adapter redirects identity conditioning. Arc2Face's generator weights remain unchanged.

Deployment:

Redirecting to a real person can transfer privacy and misuse risks to that person. Deployment should investigate synthetic targets or targets used with consent.

The Chosen redirection target matters - the least similar of the candidates improves "forgetting" with retention essentially unchanged

References:

- [1] Z. Li et al., "Photomaker: Customizing realistic human photos via stacked id embedding." CVPR, 2024.
- [2] Q. Wang et al., "Instantid: Zero-shot identity-preserving generation in seconds." arxiv.org/abs/2401.07519, 2024.
- [3] J. Sun et al., "SMI: Statistical membership inference for reliable unlearned model auditing." arxiv.org/abs/2602.01150, 2026.
- [4] B. Zhang et al., "Verification of machine unlearning is fragile." proceedings.mlr.press/v235/zhang24h.html, 2024.

